Product Label Analyzer

Sample Project Architecture, Probable Technologies & Development Roadmap

Goal: build an evidence-grounded system that accepts front and back product-packaging images, extracts label information, checks ingredients and claims against regulatory sources in India, the USA and Europe, retrieves scientific evidence, and produces a traceable assessment of how well packaging claims are supported.

1. High-Level System Architecture

Layer	Main Responsibility	Probable Technologies
User Interface	Upload front/back photos; display analysis and citations	React, Vite, HTML/CSS
API / Backend	Coordinate the complete analysis pipeline	Python, FastAPI, Pydantic
Image Processing	Resize, crop, rotate, perspective correction, enhancement	OpenCV, Pillow
OCR	Extract text from packaging images	PaddleOCR; Tesseract; Vision LLM fallback
Label Parser	Identify ingredients, nutrition, claims, allergens, manufacturer details	Python, Pydantic, LLM-assisted structured extraction
Ingredient Normalization	Map aliases, E-numbers/INS codes and names to canonical ingredients	Rules + mapping + database + embeddings
Regulatory Retrieval	Compare ingredients/claims with official country requirements	FSSAI, FDA, European Commission/EU legislation, EFSA material
Scientific Retrieval	Find relevant papers, reviews and reports	PubMed, Crossref, Google Scholar/other scholarly indexes
RAG / Evidence Engine	Hybrid retrieval, ranking, citation grounding	BM25, Sentence Transformers, FAISS, reranker
Reasoning	Synthesize retrieved evidence without inventing unsupported conclusions	LLM API + structured prompts
Database	Persist products, ingredients, regulations, sources and analyses	PostgreSQL, SQLAlchemy
Reporting	Produce structured human-readable result	React UI; optional PDF report generation

2. Core Processing Pipeline

1. Upload Front + Back Images
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2. Image Quality Check & Preprocessing
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3. OCR / Vision Extraction
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4. Structured Label Parsing
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5. Ingredient & Claim Normalization
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6. Regulatory Evidence Retrieval — India / USA / Europe
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7. Scientific Literature & Report Retrieval
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8. Evidence Ranking + RAG
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9. Packaging-Claim Assessment
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10. Grounded LLM Synthesis
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11. Citation / Evidence Validation
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12. Final Product Report

3. Detailed Architecture

3.1 Image Intake

The user supplies two images: a front image containing branding and marketing claims, and a back image containing ingredients, nutrition information, allergens and regulatory details. The system should retain the original images for traceability.

3.2 OCR and Visual Extraction

Use image preprocessing before OCR. A practical first implementation is PaddleOCR or Tesseract. A vision-language model can be used as a fallback when OCR confidence is poor or when the label has complex layouts.

3.3 Label Parser

Convert OCR text into structured fields such as product name, ingredients, nutrition facts, allergens, claims, manufacturer, country of origin, certification marks and regulatory identifiers. Pydantic models should validate the extracted structure.

3.4 Ingredient Normalization

Different labels may use aliases, E-numbers, INS numbers, abbreviations or common names. Normalize these into a canonical ingredient identity before performing searches. This prevents separate searches for equivalent names.

3.5 Regulatory Layer

The regulatory layer should prioritize authoritative material. For India, FSSAI regulations cover food products, additives, labelling/display, advertising/claims and related areas. For the USA, use FDA material. For Europe, use European Commission legislation and relevant EFSA scientific assessments. Store jurisdiction, rule, condition of use, source URL, date and relevant text with every regulatory finding.

3.6 Scientific Literature Layer

Search for systematic reviews, meta-analyses, human studies, observational studies, animal studies and other relevant evidence. Evidence quality and study type should be stored rather than treating every search result as equally reliable.

3.7 Hybrid RAG

Use keyword retrieval such as BM25 together with semantic embeddings. Combine and rerank the results, then provide only the strongest relevant evidence to the LLM. A refusal or UNABLE TO VERIFY state should be available when evidence is insufficient.

3.8 Packaging Claim Analyzer

Extract claims such as 'natural', 'high protein', 'no added sugar', 'immunity booster', 'organic' or other health/quality claims. Each claim should be assessed separately against the label, applicable rules and scientific evidence.

3.9 Evidence-Grounded Reasoning

The LLM should synthesize evidence rather than act as the source of truth. Every important conclusion should link back to a regulatory document, research paper, report or label evidence. Unsupported claims should be marked as uncertain rather than guessed.

4. Recommended GitHub Repository Structure

product-label-analyzer/

■■■ README.md

■■■ ARCHITECTURE.md

■■■ requirements.txt

■■■ .env.example

■■■ docker-compose.yml

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■■■ app/

■ ■■■ main.py

■ ■■■ api/

■ ■ ■■■ routes_upload.py
■ ■ ■■■ routes_analysis.py
■ ■ ■■■ routes_reports.py
■ ■■■ models/
■ ■ ■■■ product.py
■ ■ ■■■ ingredient.py
■ ■ ■■■ regulation.py
■ ■ ■■■ evidence.py
■ ■ ■■■ analysis.py
■ ■■■ services/
■ ■ ■■■ image_processor.py
■ ■ ■■■ ocr_service.py
■ ■ ■■■ label_parser.py
■ ■ ■■■ ingredient_normalizer.py
■ ■ ■■■ regulatory_checker.py
■ ■ ■■■ literature_search.py
■ ■ ■■■ claim_analyzer.py
■ ■ ■■■ final_analyzer.py
■ ■■■ rag/
■ ■ ■■■ retriever.py
■ ■ ■■■ reranker.py
■ ■ ■■■ embeddings.py
■ ■ ■■■ citation_resolver.py
■ ■■■ sources/
■ ■■■ fssai.py
■ ■■■ fda.py
■ ■■■ eu.py
■ ■■■ pubmed.py
■ ■■■ crossref.py
■
■■■ frontend/
■ ■■■ src/
■ ■■■ components/
■ ■■■ pages/
■ ■■■ services/
■
■■■ data/
■ ■■■ raw/
■ ■■■ processed/
■ ■■■ corpus/
■
■■■ scripts/
■ ■■■ ingest_fssai.py
■ ■■■ ingest_fda.py
■ ■■■ ingest_eu.py

■ ■■■ ingest_literature.py

■

■■■ tests/

■■■ test_ocr.py

■■■ test_parser.py

■■■ test_normalizer.py

■■■ test_regulations.py

■■■ test_retrieval.py

■■■ test_end_to_end.py

5. Probable Technology Stack

Area	Recommended Starting Choice	Alternative / Later
Language	Python	—
Backend	FastAPI	Flask
Data validation	Pydantic	Dataclasses
Image processing	OpenCV + Pillow	—
OCR	PaddleOCR	Tesseract / Vision LLM
Frontend	React + Vite	Streamlit for early prototype
Database	PostgreSQL	SQLite for MVP
ORM	SQLAlchemy	SQLModel
Keyword retrieval	BM25	Elasticsearch/OpenSearch
Embeddings	Sentence Transformers	API embeddings
Vector search	FAISS	pgvector / Qdrant
Reranking	Cross-encoder	API reranker
Literature	PubMed + Crossref + scholarly indexes	Additional APIs
Regulatory data	Official FSSAI/FDA/EU sources	Curated local corpus
LLM	API-based model	Local open-source model
Testing	pytest	—
Deployment	Docker + Docker Compose	Cloud deployment

6. Step-by-Step Development Roadmap

Phase	Build	Output
1	Project skeleton + FastAPI + basic UI	Working upload endpoint
2	Image preprocessing + OCR	Raw front/back text
3	Pydantic label schema + parser	Structured product data
4	Ingredient normalization	Canonical ingredient list
5	India regulatory module	FSSAI-backed checks
6	USA + Europe regulatory modules	Three-jurisdiction comparison
7	Scientific literature retrieval	Relevant papers/reports
8	BM25 + embeddings + reranking	Evidence-grounded retrieval
9	Claim extraction + assessment	Claim-by-claim evaluation
10	LLM synthesis + citations + abstention	Traceable final analysis
11	Database + caching + evaluation tests	Reproducible system
12	Polish frontend + Docker + documentation	GitHub-ready project

7. Development Flow Chart

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Upload Front + Back Product Images

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Validate Image Quality

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Preprocess Images

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OCR / Vision Extraction

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Parse Product Label

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Normalize Ingredients + Extract Claims

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• Clickable/source-level citations so every important conclusion can be audited.

9. Important Design Principle

Do not build a black-box 'healthy/unhealthy' classifier. The system should be an evidence-grounded analysis tool. Regulatory status, scientific evidence and packaging claims are different dimensions and should be reported separately. The LLM should synthesize retrieved evidence, not invent regulatory rules or scientific conclusions. If authoritative evidence cannot be found, the system should explicitly say that the claim is unverified.

10. Regulatory Source Note

For India, FSSAI currently publishes regulations covering food products and additives, labelling/display, advertising and claims, packaging and related subjects. FSSAI also states that its compendia are reference documents and that original Gazette notifications/amendments prevail where there is a conflict. Therefore the production system should retain the exact source document/version and retrieval date for regulatory evidence.

Sources consulted for this sample architecture: FSSAI Food Safety and Standards Regulations and FSSAI Labelling and Display Regulations. See the project documentation for the official source links and for the corresponding USA/EU regulatory sources before implementing the production regulatory corpus.